

SEAN JAFRI

Robotics Hardware Engineer • Mechanical Design • Test Automation Hardware

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SUMMARY

Robotics hardware engineer with 8+ years designing, prototyping, and deploying the robotic fixtures and automation hardware that HP Inc.'s Commercial Systems validation runs on. End-to-end ownership from concept through factory floor: SolidWorks CAD with GD&T, Eagle PCB layout, Arduino-based prototyping, MJF printing on an HP Multi Jet Fusion 4200 I operate and maintain, and direct deployment with test technicians. Track record: inherited flagship fixture redesigned across three generations for ~89% cost reduction; robot controller system cost cut ~\$800 → under \$300 per fixture.

CORE SKILLS

- Mechanical Design: SolidWorks (8+ yrs daily), GD&T per ASME Y14.5, DFM/DFA, mechanism & linkage design, tolerance stack-ups
- Fabrication: HP Multi Jet Fusion 4200 (operate & maintain — PM kits, transmission assemblies), FDM/SLA, 80/20 extrusion, manual machining, vendor CNC
- Mechatronics: servo, stepper & linear actuators; solenoids & drivers; sensor integration; Arduino-based prototyping; PCB layout (Autodesk EAGLE → migrating to Fusion 360 Electronics)
- Test Automation Hardware: design & deployment of robotic fixtures and validation infrastructure used by test engineering teams; design-for-test; high-cycle durability; root-cause analysis on fixture hardware; operator documentation (user guides + video walkthroughs)

PROFESSIONAL EXPERIENCE

Robotics Hardware Engineer (Contract via Mphasis) — HP Inc., Houston, TX Apr 2018 – Present

- Lead mechanical design and deployment of 14+ robotic test fixtures used daily to validate laptops, docks, and accessories at production scale; dock-fixture fleet of 12+ serialized stations across four labs, maintained as versioned hardware.
- Inherited flagship laptop fixture (~\$3,500 as deployed) and redesigned it across three generations to ~\$400 (~89% reduction); setup time 10 min → 2 min; deployment density doubled per bench.
- Replaced \$250–\$350 per-model FDM laptop cradles (31 produced across ~34 platforms) with a \$25 set of MJF-printed universal corner brackets — eliminating per-model retooling across the validation fleet.
- Designed Mega II controller: Eagle-routed shield over Arduino Mega 2560 driving 12 actuators with per-actuator position feedback and touchscreen UI; controller cost \$300 → \$100; controller-plus-bot hardware ~\$800 → under \$300 per fixture (firmware developed by team's software developer).
- Own end-to-end product cycle — CAD/GD&T, additive & machined prototyping, electronics integration, vendor management, BOM & inventory systems — with every fixture shipping alongside a written user guide and video walkthrough.
- Mentored a teammate in Eagle PCB design through execution of a production board revision.

SELECTED PROJECTS

- Deep Purple Dock Fixture — 12-actuator robotic dock-cycler exercising USB-C, USB-A, HDMI & DisplayPort; per-actuator position feedback; reconfigurable 8/9/12-port modules; custom Eagle control PCB; DB0 → DB1 dock-form-factor redesign.
- Robotic Laptop Fixture (3 redesign generations) — optical-breadboard original → universal-cradle → 80/20 frame → off-the-shelf holder with MJF brackets; ~\$3,500 → ~\$400.
- Keyboard Bot — 120+ solenoid array pressing every key on a laptop; MJF chassis with per-key compliance; enables automated typing-fatigue, key-rollover, and BIOS hotkey validation.
- Mega II Robotic Test Controller — 12-channel Eagle shield + touchscreen; tripled channel count at a third of the cost. • Universal Laptop Cradle — \$25 MJF bracket set holding any laptop; ended per-model cradle tooling.
- Additional — USB-C Flip Bot (planetary-gear plug rotator), Multiport Bot (3-port concurrent array: USB-C/USB-A/HDMI in any combination), Low Power Switch (4-channel modular USB enable/disable — mechanical & enclosure design), Varda Reviver (USB laptop-revive device, Eagle PCB, 255 units built).

EDUCATION & LEADERSHIP

- B.S. Mechanical Engineering — The University of Texas at Arlington, May 2017
- President / VP / Team Lead, UTA Mars Rover Team (2014–2017) — led 40+ members across three University Rover Challenge cycles (team competed 2014, 2015, 2017; on-site at URC 2015 & 2017); secured \$19,000+ in university grants and program funding, plus in-kind sponsor support (Bell Helicopter, Line-X, Protocase).
- UVS Certificate Course — six-wheeled ground vehicle taken from remote control to end-of-course autonomous navigation (GPS, ultrasonic, camera vision); focus: mechanical/chassis.
- Senior Capstone (CAD Design Lead) — Sundog active sun-tracking device for UTARI.